

Neoplasia and Antithrombotic Drugs

Study handout — Week 6. Department of Pathology and Pharmacology.

Nomenclature

A neoplasm is an abnormal mass of tissue whose growth exceeds that of normal tissue and persists after the stimulus ceases. Benign tumors are designated by attaching the suffix -oma to the cell of origin. Malignant tumors arising in mesenchymal tissue are called sarcomas.

Carcinomas are malignant neoplasms of epithelial cell origin. They are the most common malignant tumors in adults.

Features of benign and malignant tumors

Feature	Benign	Malignant
Differentiation	Well differentiated	Variable, may be anaplastic
Rate of growth	Usually slow	Often rapid
Local invasion	Absent; usually encapsulated	Present
Metastasis	Absent	Frequently present

Metastasis is the most reliable feature that distinguishes a malignant from a benign tumor. Carcinomas typically spread through lymphatics, whereas sarcomas typically spread through the blood.

Epithelial tumors of the colon



Figure 1: Tubular adenoma of the colon

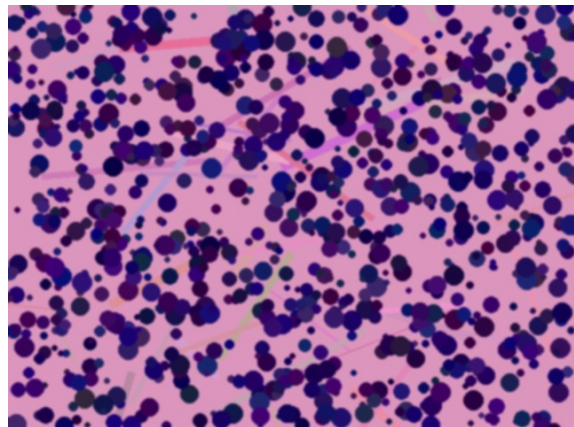


Figure 2: Invasive adenocarcinoma of the colon

A tubular adenoma is a benign neoplasm of the colonic epithelium. Invasive adenocarcinoma of the colon penetrates the muscularis mucosae.

Smooth muscle tumors of the uterus

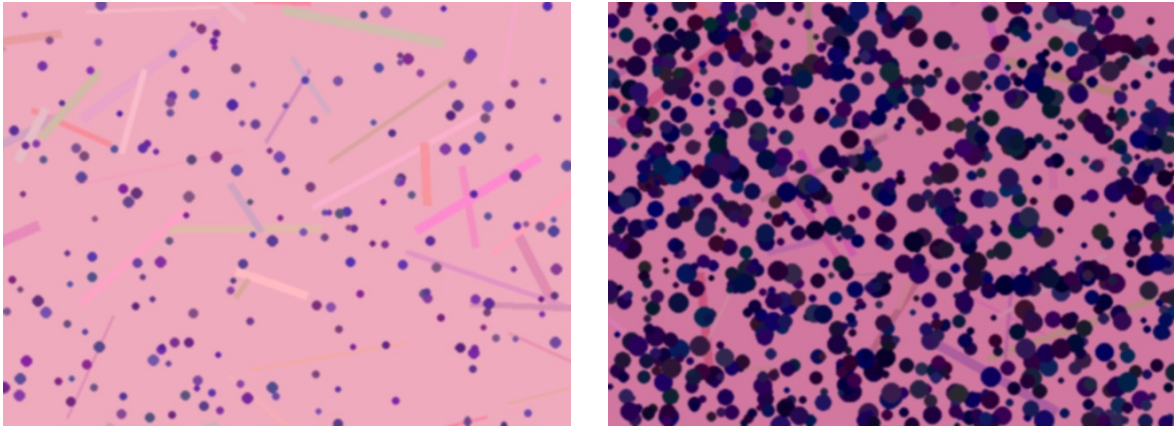


Figure 3: Leiomyoma (left) and leiomyosarcoma (right) of the uterus

Leiomyoma is the most common benign tumor in women. Leiomyosarcomas arise de novo rather than from a preexisting leiomyoma.

Carcinogenesis

Tumor suppressor genes such as TP53 and RB require loss of both alleles before a cell is transformed. Oncogenes such as RAS promote growth when a single allele is mutated.

Check your understanding: which tumor spreads mainly by lymphatics?

Anticoagulants

Heparin

Heparin activates antithrombin, which inactivates thrombin and factor Xa. It is monitored with the activated partial thromboplastin time. Its effect is reversed by protamine sulfate. It can cause heparin-induced thrombocytopenia.

Low-molecular-weight heparins such as enoxaparin act mainly on factor Xa.

Warfarin

Warfarin inhibits vitamin K epoxide reductase. It is monitored with the prothrombin time and the international normalized ratio. Its effect is reversed by vitamin K and fresh frozen plasma. It is contraindicated in pregnancy because it is teratogenic.

Direct oral anticoagulants such as rivaroxaban inhibit factor Xa directly.

Antiplatelet drugs

- Aspirin irreversibly inhibits cyclooxygenase and reduces thromboxane A₂ synthesis.
- Clopidogrel blocks the P₂Y₁₂ ADP receptor on platelets.
- Abciximab blocks the glycoprotein IIb/IIIa receptor.

Thrombolytics

Alteplase converts plasminogen to plasmin. Its major adverse effect is bleeding.

Drug interactions to remember

Drug	Monitoring test	Antidote
Heparin	aPTT	Protamine sulfate
Warfarin	PT/INR	Vitamin K
Dabigatran	None routinely	Idarucizumab

Further reading: Kumar V et al. Robbins Basic Pathology, 10th edition, chapter 6.